

On the Waveform-to-Bit Gap in Single-Channel Blind Source Separation of Co-Frequency Signals: A Decision-Identical Evaluation Protocol and Controlled Diagnosis

Bin Nong^{a,*}, Weihong Fu^b, Zhuoyun Jiang^c

^a*Tianfu College of Southwestern University of Finance and Economics, Mianyang, Sichuan, 621000, China*

^b*School of Telecommunications Engineering, Xidian University, Xi'an, Shaanxi, 710071, China*

^c*School of Health Science and Engineering, University of Shanghai for Science and Technology, Shanghai, 200093, China*

Abstract

Single-channel blind source separation (SC-BSS) of co-frequency communication signals is trained and ranked by waveform-fidelity metrics such as scale-invariant signal-to-distortion ratio (SI-SDR), yet the downstream task is demodulation. This paper contributes a bit-level evaluation protocol for SC-BSS—an offset-compensated receiver producing compensated symbol error rate (SER) and Gray-mapped bit error rate from shared decisions—together with a differentiable soft-demodulation loss verified decision-identical to that receiver, and applies them in a controlled five-seed study on a variable-count ($K \in \{1, 2, 3\}$) co-frequency benchmark. The protocol exposes a large waveform-to-bit mismatch: a +4.1 dB SI-SDR improvement moves the compensated SER from 0.5049 to 0.5027—even +8.4 dB at −10 dB SNR buys

*Corresponding author.

Email address: nongbin@tfswufe.edu.cn (Bin Nong)

nothing—because the mixture is interference-limited at $\text{SIR} \approx 0 \text{ dB}$. The task-oriented loss is negligible in the pool (at most 0.03 percentage points in the fair additive comparison) but sign-structured by the interference order: it *helps* at $K=2$ (5/5 seeds positive in all four contrasts) and *hurts* at $K=3$ (0/5 seeds). Deprived of the waveform anchor, the loss collapses into reward hacking (-9.4 dB), delimiting where the proxy can be trusted; with the anchor, the same term repairs occupancy gating (count accuracy $0.65 \rightarrow 0.85$). A joint soft-information head fails at chance, and controlled oracle-versus-blind probes attribute the failure to burst-level synchronisation and phase-entangled joint detection. Produced under a pre-declared statistical protocol and released with full code, these results provide reusable measurement infrastructure and a concrete prescription: closing the waveform-to-bit gap requires explicit synchronisation and equalisation structure, not a better loss weight.

Keywords: Blind source separation, Co-frequency communication, Evaluation protocol, Task-oriented learning, Symbol error rate, Soft demodulation, Complex-valued neural network

1. Introduction

Problem and motivation. Single-channel blind source separation (SC-BSS) recovers multiple unknown source signals from one observed mixture [1, 2]. For co-frequency communication signals—several modulated emitters
5 overlapping in time and frequency at a single receiving antenna—deep separators have progressed along a clear axis: better architectures, trained and ranked by waveform-fidelity metrics, above all the scale-invariant signal-to-

distortion ratio (SI-SDR) [3, 4] and its improvement over the mixture (SI-SDRi). Yet separation is rarely the end goal. In spectrum monitoring, cognitive radio and electronic support, the separated waveforms are fed to a *demodulator*, and the quantity the operator pays for is the bit error rate (BER). The premise of the whole line of work is that waveform fidelity is a good proxy for bit-level recoverability. This paper tests that premise directly, finds it wanting, and then tests the obvious remedy—training the separator on a differentiable demodulation loss—with equal care.

Why the premise is fragile here. Two properties of the co-frequency setting conspire against the proxy. First, SI-SDR is scale- and phase-invariant: it does not care whether the estimated waveform lands on the constellation grid that a demodulator’s decision boundaries carve out. Second, the practically relevant operating point of a co-frequency mixture is symmetric interference: with comparable per-source amplitudes the signal-to-interference ratio is $\text{SIR} \approx 0\text{ dB}$, so the residual on any separated output is not small receiver noise but *amplitude-level leakage of the other sources*. Whether any training objective can push a separated waveform across decision boundaries under such residuals is an empirical question that, to our knowledge, has not been answered with statistically controlled experiments.

What we do. We work on a variable-count co-frequency SC-BSS benchmark ($K \in \{1, 2, 3\}$ sources; BPSK/QPSK/8PSK/16QAM; root-raised-cosine pulses; multipath fading; AWGN) inherited from our companion open-world study [5], and evaluate every model with an *offset-compensated receiver*: oracle down-conversion at each source’s true carrier, matched filtering, unit-power symbol grid, phase-only alignment, minimum-distance

decisions, and Gray-mapped BER from the same decisions (Section 3). On top of this evaluation truth we build a *differentiable soft-demodulation* loss whose receiver front-end is line-by-line identical to the evaluation receiver (Fig. 1), and insert it into the variable- K permutation-invariant training (PIT) loss (Section 4). Three studies follow. S1 quantifies the SI-SDR $\rightarrow$ SER/BER mismatch on fifteen previously trained separators. S2 trains the same slot separator under five loss configurations (five seeds each) with exact and Monte-Carlo permutation tests. S3 attempts the classical end of the task-oriented road—a joint soft-information head that bypasses the explicit waveform stage—and diagnoses its failure with a controlled synchronisation probe.

Findings.

1. **The mismatch is real and large.** +4.1 dB SI-SDRi moves the compensated SER from 0.5049 (mixture, no separation) to 0.5027 and the Gray BER from 0.2669 to 0.2651; even +8.4 dB SI-SDRi at -10 dB SNR buys nothing (a fraction-of-a-point penalty, if anything), and the flatness persists across an SIR sweep to 20 dB at fixed $K=2$ (Section 5.2). The damage localises to interference: the baseline SER jumps from 0.146 at $K=1$ to 0.561 at $K=2$ and only 0.587 at $K=3$ —one interferer closes nearly the whole demodulation gap.
2. **A faithfully aligned task loss barely moves it, and the sign is set by the interference, not the loss.** Pooled over K , the soft-SER term is practically negligible (≤ 0.17 pp BER for the replacement configuration (c); ≤ 0.03 pp for the fair additive comparison (d) vs. (b)). Decomposed by source count, however, the term *helps* at $K=2$ —the

SIR ≈ 0 dB case that motivates it (BER +0.35 pp for the replacement contrast, +0.11 pp in the fair additive one, 5/5 seeds in all four contrasts, exact permutation $p = 0.0625$)—and *hurts* at $K=3$ (SER -0.6 to -0.7 pp, 0/5 seeds in all four contrasts), the two cancelling in the pool (Section 5.3).

3. **Waveform supervision is necessary but not sufficient.** A pure soft-SER loss finds off-manifold solutions that minimise the soft receiver’s cross-entropy without separating (SI-SDRi -9.4 dB, SER *worse* than the unseparated mixture)—a reward-hacking failure that also delimits where the training proxy can be trusted (Section 5.4).
4. **The task term is not useless—it regularises the occupancy structure.** Occupancy-routed count accuracy rises from 0.65 to 0.85 and the hallucination rate drops from 0.15 to 0.05–0.06, an emergent benefit for the slot *gating* that waveform losses do not provide (a dedicated count head, at ≈ 0.9 (0.87–0.92) for all configurations, still counts better—the gain is in what the gates pass downstream, not in the count itself; Sections 5.3 and 5.4).
5. **Naive joint detection fails, informatively.** A learned per-symbol posterior head sits at the chance level, and controlled probes (same head, same data, oracle vs. blind synchronisation) point to burst-level blind carrier/phase recovery as the binding constraint at $K=1$; at $K=2$ even a frequency oracle fails to beat the mixture baseline—sync alone is not sufficient under phase entanglement (Section 5.5).

The overall picture these results support is mechanistic, not statistical-anecdotal: the residual on any separated output is interference leakage, and

what a grid-aware task loss can add on top of a waveform loss depends on the *structure* of that leakage—one interferer’s residue is structured enough
85 to nudge across decision boundaries ($K=2$), two interferers’ residue fills the grid and turns the same gradient misleading ($K=3$). Closing the gap requires explicit synchronisation and equalisation structure in the model, not a re-weighted loss. Section 6 develops this argument.

The remainder of the paper is organised as follows. Section 2 reviews
90 the literature this work connects. Section 3 defines the benchmark and the compensated receiver that serves as evaluation truth. Section 4 describes the separator, the differentiable soft-demodulation loss, and the joint head. Section 5 presents S1–S3. Section 6 gives the mechanistic discussion, Section 7 states limitations, and Section 8 concludes.

95 2. Related Work

2.1. SC-BSS for communication signals

Blind separation of co-frequency communication signals from a single sensor violates the overdetermined requirement of classical ICA [6]; classical workarounds exploit time–frequency sparsity in the underdetermined
100 case [7], and the co-frequency setting has driven a line of deep-learning approaches: convolutional separation networks [1], recurrent designs [8], attention-augmented models [9], complex-domain architectures [2], spatial-temporal fusion [10], and structured state-space backbones [11]. The setting has recently been extended to an unknown number of sources by coupling
105 separation with source counting, in radar mixtures [12] and in our companion open-world study [5], whose variable- K benchmark, slot architecture and

trained checkpoints this paper builds on. All of these works train and rank models by waveform metrics (SI-SDR, SDR, NMSE); bit-level read-outs under a compensated receiver are rare, and to our knowledge no prior work tests
 110 whether waveform improvements reach the demodulated bits under symmetric co-frequency mixtures—the question addressed here.

2.2. *Waveform metrics and their discontents*

Modern time-domain separation descends from deep clustering [13], PIT [3] and Conv-TasNet [14], with SI-SDR as the dominant training and
 115 reporting metric. That the metric is not the task is well documented in audio: the BSS_EVAL suite itself warns that energy-ratio metrics conflate artefacts with interference [15], and Le Roux et al. [4] show that SI-SDR improvements can be purchased with distortions the metric cannot see, recommending task-level evaluation. We import this scepticism to co-frequency
 120 *communication* mixtures, where the task metric is sharp and unambiguous—symbol and bit error rate under a compensated receiver—and quantify the mismatch rather than assuming it.

2.3. *Task-oriented and semantic communication*

Task-oriented communication argues that transmit and receive processing
 125 should be optimised for the consumer of the message rather than for reconstruction fidelity [16, 17], and deep learning for the physical layer has made end-to-end learned transceivers practical [18]. In speech enhancement the principle delivers: metric-oriented adversarial training and joint acoustic-model training of separation front-ends improve downstream recognition where waveform metrics plateau [19, 20]. Our work is complementary
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and, in a sense, cautionary: we apply the task-oriented principle to the *separation* stage of a receiver chain for co-frequency communication mixtures, and find that there the principle is bounded by the interference structure of the channel—the attainable bit-level gain is set by the number of interferers,
 135 not by the loss.

2.4. *Learned detection and joint demodulation*

That joint multiuser detection outperforms separate-then-detect pipelines is classical [21], and practical co-channel systems have long exploited exactly that structure: paired-carrier multiple access (PCMA) lets two transmitters
 140 share one carrier and separates them at the receiver with explicit synchronisation, channel estimation and self-interference cancellation [22], while two decades of interference-cancellation receiver design treat explicit synchronisation and equalisation as the irreducible core of co-channel demodulation [23]. Neural approximations of multiuser detectors learn the detection mapping
 145 directly [24, 25], but assume symbol-synchronous, equalised inputs and (typically) known channel state—the synchronisation problem is solved before the network sees the signal. Our S3 experiment asks whether a lightweight head attached to a separation backbone can absorb joint detection *and* burst-level synchronisation at once; the answer, with a controlled probe, is that it cannot
 150 (Section 5.5)—consistent with what those classical designs presuppose.

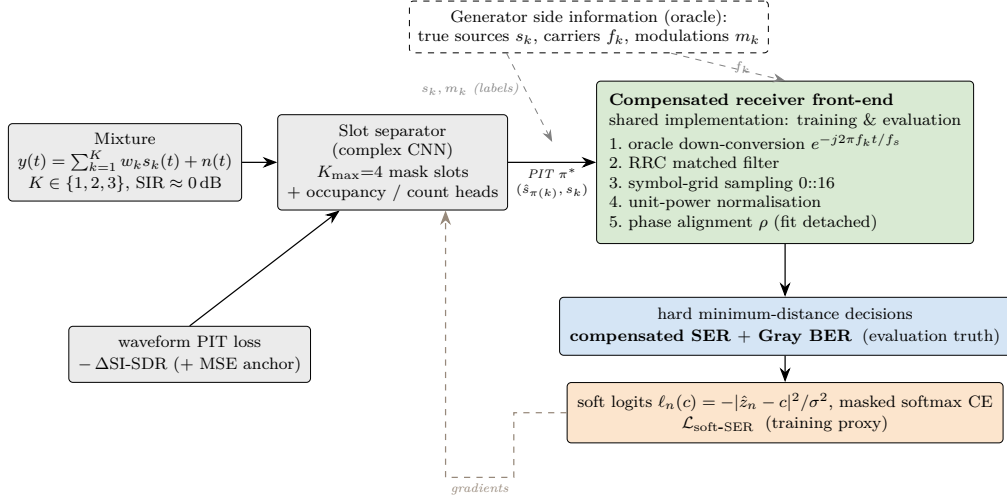


Figure 1: The task-oriented training/evaluation pipeline. A variable- K co-frequency mixture is separated into $K_{\max}=4$ slots and PIT-paired with the true sources and trained with the waveform PIT loss ($-\Delta\text{SI-SDR}$ with the MSE anchor) plus the task term. One compensated-receiver front-end (oracle down-conversion, matched filtering, symbol-grid sampling, unit-power normalisation, detached phase alignment) is shared by both sides: its hard decisions are the evaluation truth (compensated SER and Gray BER, Section 3.2–3.3), and its soft cross-entropy is the training proxy $\mathcal{L}_{\text{soft-SER}}$ (Section 4.2). Dashed grey: oracle side information exposed by the generator; dashed orange: gradients.

3. Benchmark and Evaluation Truth

3.1. Variable- K co-frequency mixtures

We observe a single complex baseband mixture of K co-frequency sources,

$$y(t) = \sum_{k=1}^K w_k s_k(t) + n(t), \quad t = 0, \dots, T-1, \quad (1)$$

with $K \in \{1, 2, 3\}$ drawn uniformly during training (a $K=4$ zero-shot probe cell exists in the test grid but is excluded from all bit-level analyses). Each source s_k is a burst of $N = 256$ symbols from a modulation alphabet $m_k \in$

{BPSK, QPSK, 8PSK, 16QAM}, pulse-shaped by a root-raised-cosine (RRC) filter (roll-off 0.35, 65 taps, a four-symbol span) at 16 samples per symbol, passed through a three-tap multipath fading channel, and carried at $f_k =$
160 $2000 \text{ Hz} + \delta_k$ with $\delta_k \sim \mathcal{U}(0, 5) \text{ Hz}$ plus an independent $\pm 5 \text{ Hz}$ jitter: the sources overlap in both time and (near-)frequency. The mixture length is $T = 4096$ samples at $f_s = 16 \text{ kHz}$. Mixing weights are drawn as $w_k \sim \mathcal{U}(0.4, 0.6)$ (unnormalised), so every multi-source cell sits at $\text{SIR} \approx 0 \text{ dB}$ by construction—the symmetric, maximally hostile case. The SNR is defined
165 on total mixture power; training SNR is uniform in $[-5, 20] \text{ dB}$ and the test grid spans $\{-10, -5, \dots, 20\} \text{ dB}$. Test sets are deterministic (seed 99999) with 100 mixtures per (SNR, K) cell. The generator is released with the code; no external dataset is required.

3.2. The offset-compensated receiver

170 Because the sources sit at *unknown* carrier offsets and the raw waveform metrics are scale-invariant, a naive demodulation proxy (nominal-carrier down-conversion, no timing or scale correction) is defective: fed the *true* 16QAM source waveform it scores SER 0.92 (0.77 with unit-power normalisation), against 0.000 for the compensated receiver (`check_naive_proxy.py`;
175 253 deterministic test sources, seed 99999). All bit-level numbers in this paper therefore come from an *offset-compensated* receiver, applied identically to the estimate $\hat{s}$ and the reference s of each matched pair:

1. **Oracle down-conversion** at the pair’s true carrier f_k (exposed by the generator): $\tilde{x}(t) = x(t) e^{-j2\pi f_k t / f_s}$;
- 180 2. **Matched filtering** with the same RRC taps (zero-delay ‘same’ convolution);

3. **Symbol-grid sampling** at offset 0 with stride 16 (the pipeline’s net symbol delay is zero);
4. **Unit-power normalisation** of each symbol stream, placing both sig-
185 nals on a common grid;
5. **Phase-only alignment** of the estimate to the reference: $\hat{z} \leftarrow \hat{z} e^{j\angle\rho}$
with $\rho = \langle z_{\text{ref}}, \hat{z} \rangle / \|\hat{z}\|^2$ fitted over all 256 symbols of the burst;
6. **Minimum-distance decisions** on the shared grid.

The compensated SER is the symbol-agreement rate of the two decision se-
190 quences; identity input scores 0.000 on all four modulations, and clean-source
decisions reproduce the transmitted symbols up to a small multipath-ISI
residual that sets the per-modulation *clean floors* (0.000/0.000/0.014/0.032
for BPSK/QPSK/8PSK/16QAM). Timing is fixed at the nominal grid (no
per-pair timing search), so the metric still measures waveform fidelity un-
195 der ideal synchronisation rather than a production receiver—an *upper-bound*
demodulation truth. This oracle assumption applies to the training loss of
Section 4.2 as well; we return to its implications in Sections 5.5 and 7.

3.3. Gray-mapped BER

The generator draws symbols directly as constellation points (no bit
200 stream), so BER is *defined* post-hoc: each decided symbol index maps to
a Gray bit string (BPSK sign bit; QPSK I/Q sign bits; binary-reflected Gray
on the 8PSK ring; per-axis Gray on 16QAM), and BER is the fraction of
mismatched bit positions between the estimate’s and the reference’s decision
bits. Formally, with $c_n^*, \hat{c}_n \in \mathcal{C}_m$ the reference and estimated decisions of

205 symbol n and $g(\cdot)$ the Gray map,

$$\text{SER} = \frac{1}{N} \sum_{n=1}^N \mathbf{1}[\hat{c}_n \neq c_n^*], \quad \text{BER} = \frac{1}{N \log_2 M} \sum_{n=1}^N d_{\text{H}}(g(\hat{c}_n), g(c_n^*)), \quad (2)$$

where d_{H} is the Hamming distance. Every nearest-neighbour pair of constellation points differs in exactly one bit (verified in the code’s smoke test). SER and BER always come from the *same* decisions.

4. Method: Task-Oriented Training

210 4.1. Slot separator

The separator is the slot-occupancy model of [5]: a lightweight complex-valued CNN (complex convolutions implemented as paired real operations [26]) with complex squeeze-and-excitation channel attention [27] encodes the mixture into a 64-channel bottleneck; a decoder produces $K_{\text{max}} = 4$
 215 complex masks, each modulating the mixture into one *slot* output; a shared occupancy head scores each slot as active/empty, and a count head predicts K directly. At inference, slots whose occupancy probability exceeds 0.5 are declared active; the occupancy route (count = number of active slots) is the counting convention used throughout this paper. The model has $\sim 254\text{K}$
 220 parameters. The architecture is inherited unchanged; the contributions of this paper are the loss, the evaluation protocol, and the analysis.

4.2. A differentiable soft-demodulation loss

The task-oriented term must satisfy one requirement above all: the training proxy and the evaluation truth must be the *same receiver*, or a null
 225 result proves nothing. We therefore implement the compensated receiver

of Section 3.2 as a differentiable torch pipeline and replace only its final hard decision by a soft cross-entropy. Concretely, for each PIT-assigned estimate/reference pair $(\hat{s}, s)$ with true carrier f_k and modulation index m , the front-end of Section 3.2 steps 1–4 produces unit-power symbol streams $\hat{z}, z \in \mathbb{C}^N$; the alignment coefficient ρ is fitted on *detached* tensors and applied as a fixed rotation $\hat{z} \leftarrow \hat{z} \rho / |\rho|$ (gradients flow through $\hat{z}$ but not through the fit, so the alignment itself cannot become a gradient exploit). Soft log-likelihoods over the per-sample constellation $\mathcal{C}_m$ are

$$\ell_n(c) = -\frac{|\hat{z}_n - c|^2}{\sigma^2}, \quad c \in \mathcal{C}_m, \quad n = 1, \dots, N, \quad (3)$$

with softness $\sigma^2 = 0.1$ on the unit-power grid, and the loss is the cross-entropy of $\text{softmax}(\ell_n)$ against the reference’s hard decision $c_n^* = \arg \min_{c \in \mathcal{C}_m} |z_n - c|$ (detached). Note that the logits are computed from the phase-aligned estimate while the labels come from the unaligned reference: the reference is the ground truth and needs no alignment—the same convention as the evaluation receiver. The total training loss is

$$\mathcal{L} = \lambda_{\text{sep}} \mathcal{L}_{\text{-SI-SDR}}^{\text{PIT}} + \lambda_{\text{occ}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{cnt}} \mathcal{L}_{\text{CE}} + \lambda_{\text{mse}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{ser}} \mathcal{L}_{\text{soft-SER}}, \quad (4)$$

where the SI-SDR term is permutation-invariant over the K true sources: with the scale-invariant signal-to-distortion ratio

$$\text{SI-SDR}(\hat{s}, s) = 10 \log_{10} \frac{\|\alpha s\|^2}{\|\alpha s - \hat{s}\|^2}, \quad \alpha = \frac{\langle \hat{s}, s \rangle}{\|s\|^2}, \quad (5)$$

(computed on zero-meaned $\hat{s}$ and s , as in the implementation), and Π_K the injective assignments of K true sources to the K_{max} slots,

$$\mathcal{L}_{\text{-SI-SDR}}^{\text{PIT}} = \min_{\pi \in \Pi_K} \frac{1}{K} \sum_{k=1}^K \left(-\text{SI-SDR}(\hat{s}_{\pi(k)}, s_k) \right), \quad (6)$$

enumerated exactly (4/12/24 assignments for $K = 1/2/3$). The MSE anchor
 245 (which fixes the output scale that pure SI-SDR leaves free) and the soft-SER
 term are computed on the *same* PIT assignment—no second enumeration—
 and $\lambda_{\text{occ}} = 1$, $\lambda_{\text{cnt}} = 0.1$ follow the companion recipe [5].

Proxy-truth parity is enforced, not assumed: on clean sources the hard
 version of the torch pipeline (argmax of Eq. (3)) reproduces the numpy eval-
 250 uation receiver’s decisions *symbol-for-symbol*, and the matched filter matches
`np.convolve` to a relative error of 3.5×10^{-7} . The identity cross-entropy is
 small but nonzero (≤ 0.45 , 16QAM worst, from the multipath-ISI residual)
 while a mismatched reference scores 11–18—the proxy thus carries a delib-
 erate gradient signal toward the grid even for a perfect waveform estimate,
 255 with a $\geq 4\times$ discriminative margin.

4.3. Training configurations

Five configurations isolate the role of each term of Eq. (4) ($\lambda_{\text{occ}} = 1$ and
 $\lambda_{\text{cnt}} = 0.1$ are always on; only the three terms below vary, and all keep the
 λ_{sep} -SI-SDR PIT term unless noted):

- 260 (a) **SI-SDR**: $\lambda_{\text{sep}} = 1$, nothing else—the paper-1-style baseline;
- (b) **MSE**: (a) + $\lambda_{\text{mse}} = 1$ —the companion study’s converged recipe [5];
- (c) **soft-SER**: (a) + $\lambda_{\text{ser}} = 1$, $\lambda_{\text{mse}} = 0$ —the task-oriented proposal; note
 that (c) *replaces* the MSE anchor, so the fair *additive* comparison is (d)
 vs. (b), not (c) vs. (b);
- 265 (d) **soft-SER + MSE**: (a) + $\lambda_{\text{ser}} = 1 + \lambda_{\text{mse}} = 1$ —combination;

(e) **pure soft-SER**: $\lambda_{\text{sep}} = 0$, $\lambda_{\text{ser}} = 1$ —no waveform objective at all (single-seed probe; checkpoint selection switches to validation soft-SER, since val SI-SDR is meaningless without the waveform term).

4.4. Joint soft-information head (S3)

270 The classical endpoint of task-oriented design is to skip the explicit waveform stage: read per-source soft symbol decisions off the shared representation directly, the neural analogue of joint multiuser detection [21]. Our head taps, per slot, the slot-masked features, the un-gated bottleneck, and the slot waveform itself ($2H+1$ complex channels), reduces frame rate to
 275 symbol rate, and emits 16-way per-symbol logits over a shared label space of capacity $M_{\text{max}}=16$, in which label j denotes the j -th point of the sample’s own constellation and a per-modulation mask activates only the first $M \in \{2, 4, 8, 16\}$ entries, trained with a constellation-masked cross-entropy against the reference hard labels on the PIT-assigned slots ($\lambda_{\text{lr}} = 1$, fixed
 280 $K=2$). Two versions were built. **v1** reduced to symbol rate by a fixed 16-sample window mean (+30.3K parameters). **v2** replaced the mean with a *learned* strided complex convolution (1-D kernel spanning two symbols, i.e. 32 samples, stride 16) that can in principle learn down-conversion and matched filtering jointly (+96.1K parameters). At inference the head’s argmax de-
 285 cisions are scored by the same SER/BER metric against the reference hard labels, alongside the waveform route through the compensated receiver.

5. Experiments

5.1. Setup and statistical protocol

Training. All S2/S3 models are the slot separator of Section 4.1 trained
290 for 100 epochs (batch 16, Adam at 10^{-3} , 2000 training and 400 validation
mixtures generated on-the-fly per epoch, gradient clipping at 1.0, and ReduceLROnPlateau on the validation SI-SDR—validation soft-SER for config
(e)), five seeds (42–46) per configuration; config (e) is a single-seed probe. S1
re-evaluates fifteen checkpoints (three architectures $\times$ five seeds) of the com-
panion study [5] with the compensated receiver—no retraining. S2’s config
305 (b) re-trains that same MSE recipe with an identical data RNG stream and
reproduces the S1 slot checkpoints exactly (bit-identical per-seed evaluation
scores), so the S1 slot row and S2’s (b) coincide by construction, not by
reuse. Evaluation uses 100 mixtures per (SNR, K) cell of the deterministic
test grid (25,600 symbols per source-matched pair cell). *Every* headline num-
ber is multi-seed; no conclusion is drawn from fewer than five seeds except
the explicitly marked single-seed probes.

Counting metrics. *Count accuracy* is the fraction of test mixtures whose
occupancy-routed count (number of slots with occupancy probability > 0.5 ,
305 floored at 1) equals the true K ; *hallucination rate* is the fraction of predicted-
active slots that cannot be matched to a true source. A separate count-head
route exists (accuracy ≈ 0.9 , range 0.87–0.92, for all configurations) but is
insensitive to the loss changes studied here; all counting numbers below use
the occupancy route.

310 *Statistics.* Means are reported with *population* standard deviation across
seeds (ddof=0). Paired comparisons use two complementary tests. (i) *Seed-*

level exact sign-flip permutation over the $n=5$ paired seed means ($2^5=32$ enumerations, two-sided): exact and assumption-light, but its smallest attainable p is $2/32 = 0.0625$ —its power is fundamentally capped, so a non-significant seed-level p is weak evidence, not proof of no effect. (ii) *Cell-level Monte-Carlo sign-flip permutation* over the $7 \text{ SNR} \times 5 \text{ seeds} = 35$ paired cell means (10^6 random sign flips): much higher power, at the price of assuming exchangeable, symmetrically distributed cell differences within a seed; a seed-level sign-agreement check accompanies every cell-level claim. Eight pre-declared contrasts (four config pairs $\times$ SER/BER) are controlled at Bonferroni $\alpha = 0.05/8$. *Sign convention:* for a contrast “X vs. Y”, $\Delta = \text{metric}(Y) - \text{metric}(X)$, so $\Delta > 0$ favours the first-named configuration X.

5.2. S1: The waveform-bit mismatch, quantified

Fig. 2 plots, for every (run $\times$ SNR) cell of the fifteen companion-study separators, SI-SDR improvement against compensated SER and BER. The cloud is vertical: SI-SDRi spans 1–9 dB across cells while SER stays pinned to the mixture baseline. Pooled over cells, the slot architecture improves SI-SDR by $+4.07 \pm 0.10$ dB yet moves SER from 0.5049 to 0.5027 ± 0.0007 and BER from 0.2669 to 0.2651 ± 0.0005 (the specialist bank behaves identically: $+4.13$ dB, 0.5018, 0.2647). In absolute terms the separated outputs remain interference-dominated: the slot model sits at mean SI-SDR $+0.1$ dB and SIR -1.6 dB—the residual interferer still carries $\sim 1.4\times$ the target’s own power. Fig. 3 localises the flatness: the per-SNR SI-SDRi falls monotonically from $+8.42$ dB at -10 dB SNR to $+1.30$ dB at 20 dB, and at every point the separated models’ SER/BER bands hug the unseparated mixture’s—an 8.4 dB

waveform gain at -10 dB SNR, the largest on the grid and a cell below the training SNR range (Section 7), buys no bit-level improvement at all (if anything a fraction-of-a-point penalty: slot SER 0.590 vs. baseline 0.583).

340 Within each SNR cell, the ten slot/specialist runs spread over 0.16–0.47 dB of SI-SDRi with SER spread ≈ 0.014 ; the per-cell correlation between SI-SDRi and SER is in fact mostly in the “correct” direction (four of seven cells with $r < -0.6$, i.e. better separation, lower SER), but the channel is practically inert: the entire within-cell SI-SDRi spread buys at most ~ 1.4 pp

345 of SER. Two per- K readings pin down where the damage lives. First, the mixture baseline itself jumps from SER 0.146 at $K=1$ (noise and multipath ISI only) to 0.561 at $K=2$ —a $3.8\times$ increase from a single interferer—and merely 0.587 at $K=3$: interference, not noise, closes the demodulation gap, and the first interferer does nearly all of it—the baseline already sits near

350 the saturated separation-failure level at $K=2$, so $K=3$ is not proportionally harder. Second, the separated models hug those per- K baselines (e.g. 0.151 vs. 0.146 at $K=1$), so separation neither repairs nor further harms the bit stream at any K .

One apparent exception is worth naming precisely because it is an arte-

355 fact: the recursive architecture’s SER of 0.4477 ± 0.0063 *looks* better than the baseline, but its 37% miss rate means SER is computed only on source pairs it managed to separate—a selection-biased subset. A matched-pair control settles this directly: on exactly the source pairs the recursive model matched, the unprocessed mixture itself scores 0.4454 ± 0.0062 —statistically indistin-

360 guishable from the model’s own 0.4477 (the model is, if anything, 0.002 *worse* on its own matched subset, and the two track each other at every SNR). The

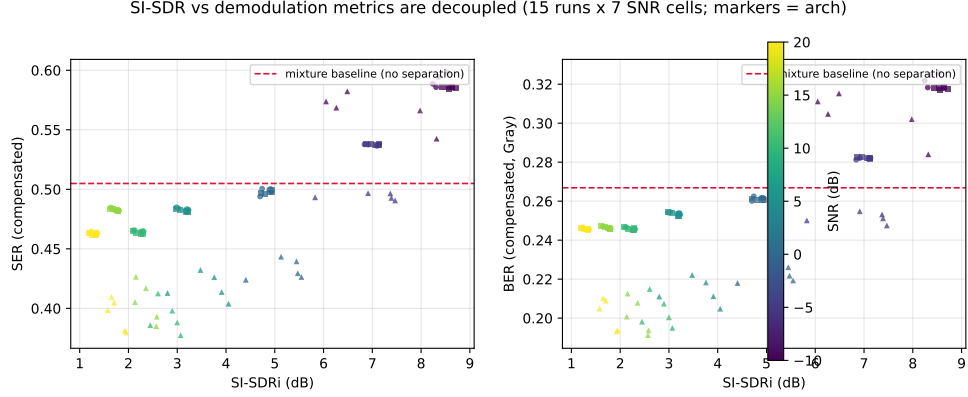


Figure 2: S1: SI-SDR improvement is decoupled from demodulation. Each marker is a (run $\times$ SNR) cell of fifteen variable- K separators (three architectures, five seeds; markers = architecture, colour = SNR); the dashed line is the unseparated-mixture baseline. SER (left) and Gray BER (right) stay on the baseline across a 1–9 dB SI-SDRi range. The recursive architecture’s sub-baseline triangle cells are a selection-bias artefact (its 37% miss rate restricts scoring to matched pairs); the honest slot/specialist cells sit on the baseline.

apparent +5.7 pp gain over the full-cohort baseline is entirely the selection effect, so the number is excluded from the comparison (it is retained in the figures, flagged, for transparency). The honest reading of S1 is that *SI-SDR-optimal separation is BER-blind at SIR ≈ 0 dB*: the residual that remains after separation is co-channel interference leakage comparable in power to the target symbol itself—several times the decision half-distance of even the densest constellation—and waveform polishing within this model class has so far not crossed the decision boundaries.

An SIR sweep at fixed $K=2$. The flatness could in principle be specific to the symmetric point $\text{SIR} \approx 0$ dB, so we re-evaluate the slot checkpoints (config (b), five seeds) on fixed- $K=2$ test sets whose mixing weights are

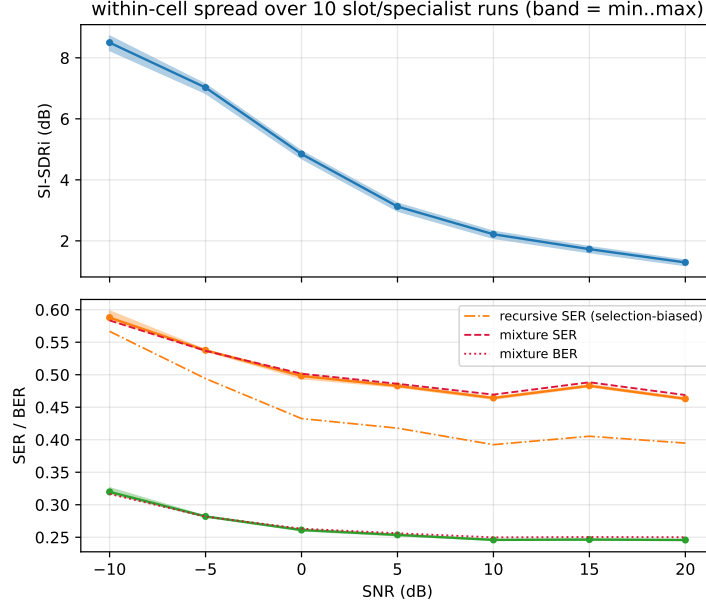


Figure 3: S1: the flat region, per SNR. Top: SI-SDRi of the slot/specialist runs (band = min..max over ten runs). Bottom: compensated SER/BER of the same runs against the mixture baselines (dashed/dotted); the recursive architecture (dash-dot) is shown separately because its miss rate biases its SER downward. Waveform quality improves by 2–8 dB across the grid while bit-level metrics do not move.

set to $\text{SIR} \in \{0, 5, 10, 20\}$ dB (geometrically symmetric about the training mean 0.5; independent test seed 88888), keeping the full SNR grid (Table 1).

375 The pooled SER *does* fall with SIR ($0.563 \rightarrow 0.466$)—but so does the per-cell mixture baseline ($0.571 \rightarrow 0.457$), and the gap between the two never exceeds one percentage point, crosses zero near $\text{SIR} = 10$ dB and turns clearly negative at 20 dB, while the SI-SDRi holds at +3.0 to +3.9 dB throughout. The SIR-driven improvement is entirely the strong source’s (SER $0.43 \rightarrow$
 380 0.19 as its interferer fades) while the weak source degrades ($0.67 \rightarrow 0.74$): asymmetric SIR does not make the *pair* more separable in bits—it merely re-

weights the cell toward the source that needed no separation. The waveform–bit mismatch is therefore not a property of the symmetric point; it persists across the swept range. (Caveat: training used $\text{SIR} \approx 0$ dB, so the $\text{SIR} > 3.5$ dB cells are out-of-distribution; the held SI-SDRi shows the separator still functions there. An SIR-matched training sweep is future work.)

Table 1: SIR sweep at fixed $K=2$ (mixing weights set to geometrically symmetric SIR levels about the training mean; independent test seed 88888, full SNR grid; config (b), 5 seeds). Δ is baseline – model in percentage points. The pooled SER falls with SIR, but so does the per-cell mixture baseline, and the gap never exceeds one point (crossing zero near 10 dB, clearly negative at 20 dB) while SI-SDRi holds — the improvement is entirely the strong source’s ($\text{SER } 0.43 \rightarrow 0.19$) as its interferer fades; the weak source degrades ($0.67 \rightarrow 0.74$). The waveform–bit mismatch is not specific to $\text{SIR} \approx 0$ dB. Models are trained at $\text{SIR} \approx 0$ dB, so the $\text{SIR} > 3.5$ dB rows are out-of-distribution.

SIR (dB)	Baseline SER	(b) SER	Δ (pp)	SI-SDRi (dB)	SER strong / weak
0	0.5710	0.5629 ± 0.0016	+0.82	+3.2	– / –
5	0.5500	0.5478 ± 0.0006	+0.22	+3.6	0.426 / 0.669
10	0.4921	0.4923 ± 0.0002	–0.02	+3.9	0.273 / 0.712
20	0.4569	0.4658 ± 0.0033	–0.89	+3.0	0.193 / 0.738

5.3. S2: Task-oriented training

Table 2 is the main matrix: five loss configurations of Eq. (4), five seeds each, evaluated by the compensated receiver. Fig. 4 shows the same numbers in the SI-SDRi $\times$ SER/BER plane. Four readings, in decreasing order of what a reader might hope for.

Table 2: S2 main matrix (slot architecture, $K \in \{1, 2, 3\}$, 5 seeds except where noted; mean $\pm$ population std across seeds). The soft-demodulation loss has at most a practically negligible effect on bit-level metrics. Seed-level exact paired permutation tests ($n=5$, two-sided; smallest attainable $p=2/2^5=0.0625$, power-limited) give (c) ser vs. (a) SI-SDR, BER: $\Delta = +0.0008$, $p=0.1875$; (c) ser vs. (b) MSE, BER: $\Delta = +0.0013$, $p=0.1250$; (d) ser+mse vs. (a) SI-SDR, SER: $\Delta = -0.0012$, $p=0.1875$. A cell-level test ($7 \text{ SNR} \times 5 \text{ seeds} = 35$ paired differences, 10^6 Monte-Carlo sign-flips) returns small p -values for config (c) ((c) vs. (a), BER: $\Delta = +0.00102$, $p=0.0004$; (c) vs. (b), BER: $\Delta = +0.00169$, $p=0.0001$, nominally surviving a Bonferroni $\alpha=0.05/8=0.00625$ over the 8 contrasts. The effect is concentrated in the high-SNR cells (15–20 dB, +0.2–0.3 pp in the cell means; 4/5 seeds agree in sign, seed-level $p=0.125$ power-capped) and vanishes at low SNR; the fair additive comparison (d) vs. (b) shows nothing (SER $p=0.38$, BER $p=0.61$), nor does any SER contrast) — but it treats within-seed cells as exchangeable even though they share one trained model, so it is anti-conservative and is reported as descriptive only; the defensible summary at $n=5$ is $|\Delta| \leq 0.17$ pp with an unstable sign. Config (c) replaces the MSE anchor with the soft-SER term; the additive design is (d). The soft-SER term does improve source counting (count acc. $0.65 \rightarrow 0.85$, occupancy route) and suppresses hallucinated sources. Configs (a)–(e) as defined in Section 4.3. [†]single seed ($n=1$), no std.

Config	SI-SDRi (dB)	SER	BER	Count acc.	Halluc.
(a) SI-SDR	$+4.23 \pm 0.02$	0.5022 ± 0.0002	0.2646 ± 0.0001	0.649 ± 0.018	0.155 ± 0.009
(b) MSE	$+4.07 \pm 0.10$	0.5027 ± 0.0007	0.2651 ± 0.0005	0.658 ± 0.023	0.149 ± 0.015
(c) soft-SER	$+3.91 \pm 0.02$	0.5031 ± 0.0017	0.2638 ± 0.0009	0.850 ± 0.088	0.060 ± 0.035
(d) soft-SER + MSE	$+3.72 \pm 0.10$	0.5034 ± 0.0013	0.2647 ± 0.0011	0.857 ± 0.032	0.049 ± 0.017
(e) pure soft-SER	$-9.42^\dagger$	$0.5159^\dagger$	$0.2713^\dagger$	$0.913^\dagger$	$0.034^\dagger$
Mixture (no separation)	–	0.5049	0.2669	–	–

1. *Pooled bit-level effect: practically negligible, and not statistically resolvable at $n=5$ seeds.* The seed-level exact test—the cluster-correct one, since the seven SNR cells of a seed share a single trained model—is power-capped ($p \geq 0.0625$ attainable) and finds nothing: (c) vs. (a) BER $\Delta = +0.0008$,

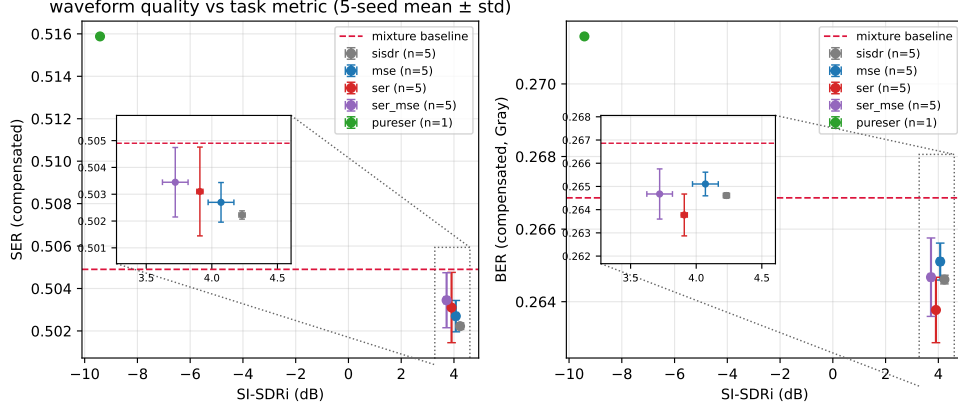


Figure 4: S2: waveform quality vs. task metrics (5-seed mean $\pm$ population std). Configs (a)–(d) cluster within 0.5 dB SI-SDRi and 0.001 SER of each other, all on the mixture baseline (dashed); the pure task loss (e) collapses off the manifold (left edge).

$p = 0.1875$; (c) vs. (b) BER $\Delta = +0.0013$, $p = 0.1250$; (d) vs. (a) SER $\Delta = -0.0012$, $p = 0.1875$. The cell-level test pools 35 (seed $\times$ SNR) differences and does return small p -values for config (c)—BER $\Delta = +0.00102$ vs. (a) ($p = 0.0004$) and $\Delta = +0.00169$ vs. (b) ($p = 0.0001$), both surviving Bonferroni over the eight contrasts, concentrated in the high-SNR cells (15/20 dB: +0.2–0.3 pp in the cell means, 4/5 seeds agree in sign)—but its sign-flips treat within-seed cells as exchangeable, which they are not; the test is anti-conservative here and we report it only as descriptive. The defensible summary at $n=5$ is $|\Delta| \leq 0.17$ pp with an unstable sign, not a demonstrated effect. But (c) *replaces* the MSE anchor (Section 4.3), so these numbers confound the task term with the anchor’s removal; the fair additive comparison is (d) vs. (b), which buys at most ~ 0.03 pp (BER $\Delta = +0.00027$, $p = 0.61$) and is null on SER ($p = 0.38$). One marginal contrast deserves disclosure: (d) vs. (a) SER has cell-level $\Delta = -0.00173$ with permutation $p = 0.064$ —

410 non-significant under our primary test—yet a paired-bootstrap 95 % CI of $[-0.00377, -0.00005]$ grazing zero on the unfavourable side. Percentile bootstrap CIs are anti-conservative at $n=5$ (this is why the permutation test is primary here), and the per- K decomposition below shows where this negative tail comes from.

415 2. *The pool hides a sign flip along the interference axis.* Table 3 decomposes the four contrasts by source count (seed-level exact permutation per K). At $K=1$ —no interference, receiver-limited—the contrasts are small and sign-unstable ($-0.01/-0.19/+0.26/+0.45$ pp BER across the four contrasts; at best 4/5 seeds agree, and the largest value is driven mostly by a single
420 seed). At $K=2$, the SIR ≈ 0 dB case that motivates task-oriented training, the term *helps*, consistently: BER $+0.35$ pp (c vs. a) / $+0.40$ pp (c vs. b) and $+0.11$ pp / $+0.06$ pp in the fair additive contrasts (d vs. b / d vs. a), with 5/5 seeds positive in *all four* contrasts (exact $p = 0.0625$, the $n=5$ floor). At $K=3$ the sign reverses: SER -0.60 to -0.72 pp with 0/5 seeds positive in
425 *all four* contrasts ($p = 0.0625$). Weighted by the 1:2:3 pair counts, the $K=2$ gain and the $K=3$ loss cancel into the pooled ≤ 0.17 pp of Table 2. The number of interferers is a more incisive experimental variable than the SNR proxy of reading 1: same loss, same backbone, same SNR grid—and the sign of the bit-level effect flips with the interference order. The negative tail of
430 the (d) vs. (a) SER contrast above is precisely the $K=3$ effect.

Table 3: S2 per- K decomposition of the four pre-declared contrasts (5 paired seeds per cell; exact two-sided sign-flip permutation, $2^5=32$). Cells give the mean paired difference in **percentage points** (positive favours the first-named configuration) and the number of seeds with a positive difference. Exact p : 0.0625 for 5/5 or 0/5 (the $n=5$ floor), 0.125 for 1/5, ≥ 0.125 for 4/5 (the exact value depends on the seed magnitudes), ≥ 0.25 otherwise. The pooled null of Table 2 resolves into structure: at $K=2$ the soft-SER term helps (BER positive, 5/5 seeds in *all four* contrasts); at $K=3$ it hurts (SER negative, 0/5 seeds in *all four* contrasts); at $K=1$ there is no interference and the contrasts are small and sign-unstable. The two cancel under the 1:2:3 pair-count weighting of the pool.

Contrast	Metric	$K=1$	$K=2$	$K=3$
(c) soft-SER vs (a) SI-SDR	SER	−0.03 (2/5)	+0.53 (4/5)	−0.72 (0/5)
(c) soft-SER vs (a) SI-SDR	BER	−0.01 (3/5)	+0.35 (5/5)	−0.15 (0/5)
(c) soft-SER vs (b) MSE	SER	−0.44 (2/5)	+0.59 (3/5)	−0.71 (0/5)
(c) soft-SER vs (b) MSE	BER	−0.19 (2/5)	+0.40 (5/5)	−0.14 (0/5)
(d) soft-SER+MSE vs (b) MSE	SER	+0.53 (4/5)	+0.32 (4/5)	−0.60 (0/5)
(d) soft-SER+MSE vs (b) MSE	BER	+0.26 (4/5)	+0.11 (5/5)	−0.10 (1/5)
(d) soft-SER+MSE vs (a) SI-SDR	SER	+0.94 (4/5)	+0.25 (4/5)	−0.61 (0/5)
(d) soft-SER+MSE vs (a) SI-SDR	BER	+0.45 (4/5)	+0.06 (5/5)	−0.12 (1/5)

3. *A structural side benefit: occupancy gating, with a caveat.* The soft-SER term markedly improves the occupancy structure: occupancy-routed count accuracy rises from 0.649 ± 0.018 (a) / 0.658 ± 0.023 (b) to 0.850 ± 0.088 (c) / 0.857 ± 0.032 (d), and the hallucination rate falls from ~ 0.15 to 0.05 –
435 0.06 ; the effect is largest exactly where the occupancy route was broken: its $K=2$ count accuracy rises from a 0.035 mean (per-seed 0.001–0.11) to 0.63 (per-seed 0.19–0.84—the spread is wide, and one seed barely improves). Two qualifications keep this benefit honest. First, the occupancy route’s tol-1

count accuracy is already 0.97–1.00 in *all* configurations—it always counts
 440 within one—so the term largely *recalibrates the $K=2$ occupancy threshold*
 rather than conferring counting ability; a threshold-free gating score would
 be the cleaner evidence and is future work. Second, the dedicated count head
 sits at ≈ 0.9 (0.87–0.92) for all configurations and still beats the repaired
 occupancy route, so the task term does not produce the best K estimate.
 445 The defence: the occupancy probabilities *gate the slot outputs*—they decide
 which waveforms are passed to the downstream receiver—and hallucinated
 slots are output waveforms, not just a wrong count. Halving hallucinated
 outputs is a bit-stream-relevant improvement that the count-head number
 does not capture.

450 4. *The waveform objective is load-bearing.* Config (e), the pure task loss,
 reaches SER 0.5159—*worse than the unseparated mixture*—with SI-SDRi
 -9.42 dB, while its training soft-SER decreases normally ($2.24 \rightarrow 1.81$): the
 optimiser found waveforms that minimise the soft receiver’s cross-entropy
 without separating, an off-manifold reward hack. The proxy–truth parity
 455 verified on clean sources (Section 4.2) holds only near the clean manifold;
 far from it, proxy and truth decouple, and only the waveform terms confine
 the model to the manifold. Waveform supervision is thus *necessary but not*
sufficient.

5.4. Ablations: dose, temperature, and what is not happening

460 Table 4 varies the two knobs of the task term (single seed). Neither
 shows a dose–response on the task metrics: across $\lambda_{\text{ser}} \in \{0.3, 1, 3\}$ and $\sigma^2 \in$
 $\{0.05, 0.1, 0.2\}$, compensated SER/BER stay within ≈ 0.005 of the controls,
 while λ_{ser} monotonically trades SI-SDRi ($+4.2 \rightarrow +2.7$ dB at $\lambda_{\text{ser}} = 3$) for

occupancy accuracy ($0.68 \rightarrow 0.93$) and hallucination suppression ($0.14 \rightarrow$
465 0.02). The counting benefit is therefore real, monotone, and cheap; the bit
benefit is not purchasable at any dose. Consistently, the training trajectory
of the soft-SER term in config (c) is nearly flat ($2.95 \rightarrow 2.84$): at $\lambda_{\text{ser}} = 1$
its gradient is dominated by the waveform terms, yet raising its weight a full
order of magnitude still buys no SER/BER movement—the flatness is not
470 an optimisation artefact of loss scale.

Table 4: S2 ablation of the soft-demodulation loss weight λ_{ser} and the soft-decision Gaussian width σ^2 (single seed 42, slot architecture). Neither knob shows a dose-response on compensated SER/BER — all values remain within ≈ 0.005 of the controls — while larger λ_{ser} monotonically improves occupancy accuracy and reduces hallucination, at the cost of SI-SDRi. Counting metrics use the occupancy route.

Config	λ_{ser}	σ^2	SI-SDRi (dB)	SER	BER	Occ. acc.	Halluc.
sisdr (control)	—	—	+4.26	0.5020	0.2644	0.684	0.139
ser (control)	1	0.1	+3.90	0.5018	0.2631	0.912	0.040
ser, $\lambda_{\text{ser}}=0.3$	0.3	0.1	+4.19	0.5029	0.2643	0.768	0.105
ser, $\lambda_{\text{ser}}=3.0$	3	0.1	+2.67	0.5046	0.2647	0.932	0.023
ser, $\sigma^2=0.05$	1	0.05	+3.16	0.5054	0.2655	0.922	0.028
ser, $\sigma^2=0.2$	1	0.2	+4.13	0.5033	0.2644	0.831	0.077

Two negative checks complete the picture. *No implicit equalisation*: config (c)’s per-modulation SER (0.236, 0.412, 0.620, 0.742 for BPSK, QPSK, 8PSK, 16QAM) remains far from the clean-source floors (0.000, 0.000, 0.014, 0.032), so the task loss did not learn to invert the multipath channel, even
475 implicitly. *No hidden capacity issue*: the counting gains show the term’s gradients do reach and reshape the backbone—the bottleneck is not gradient

flow but the interference structure itself.

5.5. S3: Joint soft-information detection—a diagnosed negative result

S3 tests the strongest task-oriented design: bypass the explicit waveform and read per-source symbol posteriors off the joint representation (Section 4.4), at fixed $K=2$ where the $\text{SIR} \approx 0$ pressure is real but the count is known. Table 5 and Fig. 5 give the outcome: both head versions sit at the modulation-averaged chance level ($\text{SER} \approx 0.73$ vs. chance 0.766; $\text{BER} \approx 0.46$ vs. random-bit 0.500), far *worse* than the unseparated mixture (0.5612/0.2872) and than the same model’s own waveform route through the compensated receiver (0.5437/0.2758, which stays intact—SI-SDRi($K=2$) +3.46 dB). The remaining row anchors this cross-study: the S2 task-loss model’s waveform route on the same $K=2$ cell (0.5475/0.2777) sits within one std of S3’s own waveform route, so the backbone operates at its expected level when the head fails. Paired bootstrap differences are decisive ($\Delta\text{SER} +0.1860$, 95 % CI [+0.1805, +0.1915] for v2 vs. the waveform route). The classical “joint detection beats separate-then-detect” does *not* come for free by attaching a head.

Table 5: S3 joint LLR head vs. waveform routing, $K=2$ cell (5 seeds, mean $\pm$ std; baseline is single-pass). The joint head is significantly WORSE than every reference: paired bootstrap gives $\Delta\text{SER} = +0.1860$ (95% CI $[+0.1805, +0.1915]$) for v2 vs. the S3 waveform route and $\Delta\text{SER} = +0.1685$ ($[+0.1670, +0.1701]$) vs. the mixture baseline — all CIs far from 0 (v1: $\Delta\text{SER} = +0.1857$ vs. waveform). Both variants sit at the modulation-averaged chance level ($\text{SER} \approx 0.73$, $\text{BER} \approx 0.46$). Candidate failure causes: v1’s boxcar averaging cancels the nominal 2000 Hz carrier almost exactly (inferred from the architecture, not verified on features); v2’s learned strided convolutions cannot recover burst-level carrier/phase blindly — a controlled $K=1$ probe with oracle sync trains the same head to val SER 0.15 vs. 0.68 under blind sync, but at $K=2$ even a frequency-oracle probe (per-source phase alignment is impossible for a mixture) reaches only 0.589, worse than the mixture baseline (0.561): synchronisation alone is not sufficient under phase entanglement.

Route ($K=2$)	SER	BER (Gray)
Mixture (no separation)	0.5612	0.2872
S2-ser waveform route	0.5475 ± 0.0063	0.2777 ± 0.0031
S3 waveform route (v2 model)	0.5437 ± 0.0083	0.2758 ± 0.0056
S3 joint head v1 (window mean)	0.7324 ± 0.0021	0.4668 ± 0.0015
S3 joint head v2 (learned strided conv)	0.7297 ± 0.0018	0.4639 ± 0.0011

The failure is informative because it has two separately identified causes.

495 *v1 (window mean)*: the fixed 16-sample boxcar averages two full cycles of the nominal 2000 Hz carrier at $f_s = 16$ kHz, so on a passband waveform its mean cancels almost exactly—the true carriers wander within $-5/+10$ Hz of the nominal value, so the cancellation is exact only at 2000 Hz; v1’s head consumes *learned features* rather than the raw waveform, so carrier starvation

500 is inferred—from the architecture and from a training plateau in which joint 16QAM SER sits at $\approx 0.92 \approx$ chance—rather than directly verified on the features themselves. We therefore treat the window-mean trap as a candidate

S3 joint LLR head fails at $K=2$; waveform route is intact (5 seeds, mean $\pm$ std)

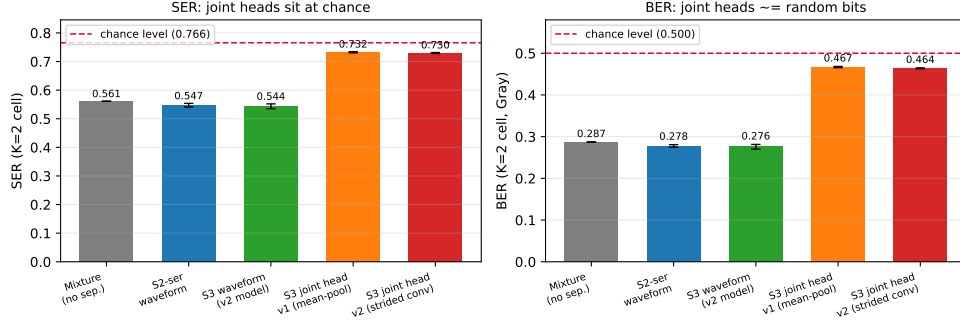


Figure 5: S3 ($K=2$ cell, 5 seeds): joint-head routes (orange/red) sit at chance (dashed), while the waveform route through the compensated receiver (green) is intact. The failure is in the head’s front-end, not in the shared backbone.

cause, not the established one. *v2 (learned strided conv)*: replacing the trap changes nothing (0.7297 vs. 0.7324)—granted, the replacement’s 32-tap kernel is itself shorter than the 65-tap matched filter, so the replacement is only a partial fix—because the deeper obstacle remains: each source still carries an arbitrary burst phase and residual frequency offset that the evaluation receiver removes with oracle knowledge and a 256-symbol phase fit, while the symbol-rate head’s receptive field spans only ± 4 symbols. A controlled probe supports this attribution: the *same* head architecture, on $K=1$ data with no separation and no PIT, plateaus at val SER 0.68 ($\approx$ chance) when it must synchronise blindly, but reaches 0.150 and is still improving when given oracle down-conversion and burst-level phase alignment (Fig. 6; per-modulation: BPSK 0.03, QPSK 0.09, 8PSK 0.25, 16QAM 0.24 vs. 0.38/0.68/0.85/0.81 blind). Note what the probe does *not* show: even with oracle synchronisation the head’s 0.150 merely matches the $K=1$ waveform-route baseline

(0.146)—the probe establishes blind synchronisation as a *necessary* explanation of the failure, not that fixing it would be *sufficient*. We read this as pointing to burst-level blind carrier/phase recovery as the binding constraint for neural joint detection in this setting; the probe is $K=1$ while the head fails at $K=2$, so we also run the $K=2$ version of the same probe (`probe_sync_head_k2.py`).

At $K=2$ the oracle can only be partial: down-converting the mixture at each source’s true carrier removes the frequency search (two oracle channels), but per-source phase alignment is impossible in principle—the two sources’ phases are entangled in the single observed mixture. With this frequency oracle the two-slot probe head reaches val SER 0.589, *worse* than the unseparated-mixture baseline (0.561) and far worse than the waveform route through the separator (0.5475); the blind variant stalls at 0.711. The $K=1$ rescue therefore does not transfer: at $K=2$ the binding constraint is not the frequency search but joint detection under phase entanglement—precisely the structure that classical multiuser detection makes explicit [21, 22].

6. Discussion

What the task loss can and cannot buy. The soft-SER term and the SI-SDR term are different metrics, but they attack overlapping feasible sets. SI-SDR is invariant to scale and to a global phase rotation, and the oracle-synchronised soft receiver removes exactly those two receiver-side nuisances (unit-power grid, phase-only alignment)—no more. It performs no timing search and no equalisation, so timing- and ISI-shaped errors lie outside the reach of *both* objectives (SI-SDR is in any case shift-sensitive, and Section 5.4

Controlled probe: burst-level blind carrier/phase sync is the bottleneck

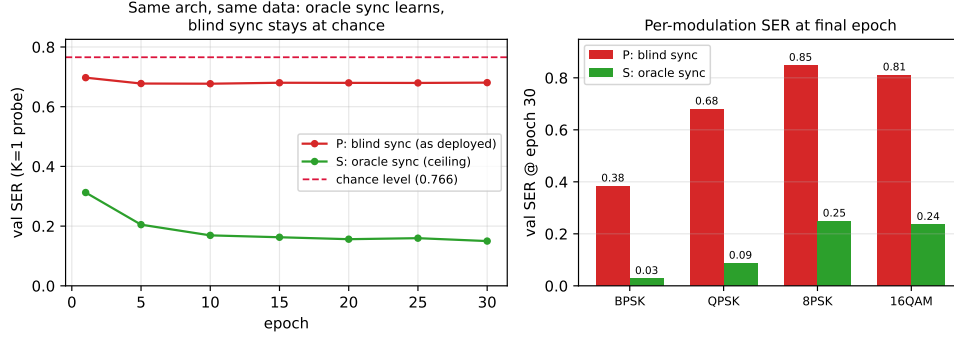


Figure 6: Controlled synchronisation probe ($K=1$, no separation): the same head learns with oracle synchronisation (green, val SER 0.31→0.15) and stalls at chance when synchronising blindly (red). Left: learning curves; right: per-modulation SER at the final epoch.

confirms that no implicit equalisation emerged). What the soft term adds over the waveform loss is therefore not a new degree of freedom but new *information*: where the residual mass sits relative to the constellation’s decision boundaries. The per- K decomposition prices that information, and it is the interference order that sets the price. At $K=1$ the residual is noise and multipath ISI; boundary information buys nothing. At $K=2$ a single leaked interferer leaves recoverable structure—borderline symbols sit close enough to a boundary that a grid-aware gradient nudges a reproducible fraction of them across (+0.35 pp BER, 5/5 seeds). At $K=3$ the same gradient turns misleading: with two interferers leaking, the residual cloud contains the *other* sources’ modulated mass as well as the target’s, and pulling estimates toward the nearest constellation points can pull them toward the wrong constellation (−0.6 to −0.7 pp SER, 0/5 seeds). We flag this $K=3$ account as a hypothesis

fitted to the observed sign pattern; the sign pattern itself—and its cancella-
 555 tion in the pool—is the robust observation. Two mundane alternatives are
 not excluded: the loss averages the soft-SER term over a sample’s sources,
 so a $K=3$ sample receives one third of the per-source task gradient of a
 $K=1$ sample (gradient dilution rather than misdirection), and at $\sigma^2 = 0.1$
 on the unit-power grid the soft receiver’s logits saturate—16QAM’s nearest-
 560 neighbour distance already gives $d^2/\sigma^2 \approx 4$, and typical residuals sit much
 further off-grid—which by itself would explain both the flat soft-SER training
 trajectory ($2.95 \rightarrow 2.84$) and the flat dose-response; the σ^2 ablation reached
 only 0.2, still inside the saturated regime. A per- K dose-response at larger
 σ^2 would separate these accounts. All effects stay in the fraction-of-a-point
 565 range because at $\text{SIR} \approx 0$ dB the leakage power is comparable to the target
 symbol’s own (several times the 0.32 decision half-distance of 16QAM on the
 unit-power grid): no loss reweighting changes the interference order. The
 dose-response ablation (Section 5.4) closes the alternative explanation that
 the term was simply under-weighted, and the additive (d) vs. (b) reading
 570 caps the deployable gain at ~ 0.03 pp.

What the soft receiver sees and what it does not. The reward hack of config
 (e) shows the proxy-truth alignment is a local guarantee: on or near the
 clean manifold the differentiable receiver *is* the evaluation receiver (symbol-
 for-symbol parity), but nothing prevents the optimiser from leaving that
 575 manifold when the waveform anchor is removed, and off-manifold the proxy
 can be minimised while true SER worsens. Task losses, like adversarially
 trained discriminators, need the generator pinned to the data manifold; here
 the waveform terms play that role.

Why counting improves. The occupancy head’s errors are leakage errors: residual energy of a true source bleeding into a nominally empty slot (hallucination) or a true slot failing to concentrate its source (missed occupancy, e.g. the $K=2$ collapse to a 0.035 mean under waveform losses). A demodulation-shaped penalty is sensitive to exactly this energy’s *position relative to decision boundaries*, not merely its total power, and it reaches the occupancy head through the shared backbone. The result—occupancy-routed counting repaired from 0.65 to 0.85 (from near-chance at $K=2$) without touching the count loss—suggests task-shaped terms as general regularisers for structured prediction heads, even where they cannot improve the task metric itself.

Consequences for system design. Three of our results point the same way. S1: waveform gains do not reach the bits at $\text{SIR} \approx 0$ —nor anywhere in an SIR sweep up to 20 dB, where the gain over the per-cell baseline never exceeds one point. S2: loss-level task terms buy a small, sign-unstable effect whose direction is dictated by the interference order—helpful against one interferer, harmful against two—not a dependable gain. S3: folding detection into the network fails at the synchronisation step that classical receivers solve explicitly and our evaluation oracle assumes away—and at $K=2$ even a frequency oracle does not rescue the head (0.589 vs. the 0.561 mixture baseline), so what is missing is not sync alone but joint detection under phase entanglement. Together they say the binding constraint is *algorithmic structure*, not objective tuning: explicit burst-level synchronisation and equalisation (carrier/phase tracking, timing recovery, ISI inversion) inside or alongside the separator, coupled with explicit joint or successive multiuser detection—or permutation-invariant training directly on multiuser-detection

likelihoods with synchronisation built in. The benchmark, the compensated
 605 receiver, and the parity-verified soft receiver released with this paper provide
 the measurement infrastructure for exactly that next step.

7. Limitations

First and most important: the compensated receiver (and hence the
 training loss) uses *oracle* carrier synchronisation and a burst-level phase fit.
 610 This is a deliberate upper-bound truth—it isolates waveform fidelity from
 synchronisation—and S3’s probe shows the oracle is doing real work that
 a deployed blind receiver would have to do itself; our SER/BER numbers
 are therefore optimistic as absolute values, though all *comparisons* share the
 same oracle. Second, all signals are synthetic (RRC pulses, three-tap multi-
 615 path, AWGN); the interference-limited mechanism should transfer, but the
 absolute numbers need not. Third, the variable- K range is $K \leq 3$ (trained)
 and the joint head was studied at fixed $K=2$ only. Fourth, the reference
 labels carry the channel’s multipath-ISI floors (up to 3.2% on 16QAM), a
 label noise shared by all configurations. Fifth, the cell-level permutation test
 620 assumes exchangeable within-seed cell differences, which does not hold (the
 seven cells of a seed share one trained model); we treat it as descriptive only
 and rely on the power-capped seed-level test throughout. Sixth, *training*
 used $\text{SIR} \approx 0$ dB throughout (source weights $\sim U(0.4, 0.6)$): the evaluation-
 time SIR sweep of Section 5.2 extends the picture to $\text{SIR} \leq 20$ dB, but cells
 625 beyond $\text{SIR} > 3.5$ dB are out-of-distribution for these checkpoints, and an
 SIR-matched *training* sweep remains future work. Seventh, the -10 dB SNR
 test cell lies below the training range ($[-5, 20]$ dB), so the $+8.4$ dB SI-SDRi

there is attained at an extrapolated operating point; we use it only as the largest-gain cell and no pooled claim depends on it. Eighth, the $K=2$ synchronisation probe uses a frequency-only oracle (per-source phase alignment is impossible for a mixture) and the same small probe head trained for 30 epochs; a larger head or longer training might narrow the 0.589 vs. 0.561 gap, though the learning curve has flattened. Finally, config (e) and the ablations are single-seed probes, used only where marked.

8. Conclusion

We have presented, to our knowledge, the first controlled study of the waveform-to-bit pipeline in single-channel blind separation of co-frequency communication signals. On a variable-count benchmark with an offset-compensated receiver as evaluation truth, SI-SDR improvements of +4 dB—up to +8.4 dB at -10 dB SNR—leave SER/BER on the unseparated-mixture baseline (a fraction-of-a-point penalty, if anything, at the lowest SNR), and the flatness persists across an SIR sweep to 20 dB at fixed $K=2$ (S1). A differentiable soft-demodulation loss, verified decision-identical to the evaluation receiver, has a pooled effect that is practically negligible and not statistically resolvable at five seeds (≤ 0.17 pp BER; ≤ 0.03 pp in the fair additive comparison); per source count, it helps consistently at $K=2$ and hurts consistently at $K=3$, cancelling in the pool (S2). The same term repairs occupancy gating (count accuracy $0.65 \rightarrow 0.85$, hallucination $0.15 \rightarrow 0.05$) and, deprived of the waveform anchor, collapses into reward hacking—waveform supervision is necessary but not sufficient. Folding demodulation into a joint head fails at chance; a controlled probe assigns the cause to

burst-level blind synchronisation at $K=1$, while at $K=2$ even a frequency oracle fails to beat the mixture baseline—synchronisation is necessary but not sufficient under phase-entangled interference (S3). The practically useful
655 reading: bit-level gains are *interference-limited, not loss-limited*—progress requires explicit synchronisation, equalisation, and joint-detection structure, evaluated at the bit level with the protocol released here.

CRedit authorship contribution statement

Bin Nong: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visu-
660 alization. **Weihong Fu:** Supervision, Project administration, Writing – review & editing. **Zhuoyun Jiang:** Resources, Validation, Writing – review & editing.

Declaration of competing interest

665 The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code and synthetic data generators are available at [https://github.com/BinNong/single_channel_source_seperation/tree/main/paper5_](https://github.com/BinNong/single_channel_source_seperation/tree/main/paper5_task_oriented)
670 [task_oriented](https://github.com/BinNong/single_channel_source_seperation/tree/main/paper5_task_oriented). No external datasets are required.

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675 Declaration of generative AI and AI-assisted technologies in the writing process

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